Zijian Huang

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Education

- **University of Michigan, Ann Arbor** *Aug. 2026 - May 2031 (Expected)*
  Ph.D. in Industrial & Operations Engineering Ann Arbor, United States
- **University of Michigan, Ann Arbor** *Aug. 2024 - May 2026*
  B.S.E. in Industrial & Operations Engineering, Dual Degree Program Ann Arbor, United States
  - GPA: 4.00/4.00
- **Shanghai Jiao Tong University** *Aug. 2022 - Present*
  B.Eng in Electrical and Computer Engineering, Dual Degree Program Shanghai, China
  - Average Score: 91.15/100.00 (ranking: 3/238)
  - GPA: 3.84/4.00 (ranking: 6/238)

Research Experience

- **Transfer Learning for Multi-Objective Bayesian Optimization** *Apr. 2025 - May 2026*
  Advisor: Prof. Raed Al Kontar
  - Developing ParetoTransfer, an archive-only transfer learning framework for multi-objective Bayesian optimization that uses source Pareto-set geometry to warm-start and guide target-task optimization.
- **Understanding Generalization in Diffusion Models** *Oct. 2024 - May 2026*
  Advisor: Prof. Qing Qu
  - Developing a theoretically grounded and practical metric, **Probability Flow Distance (PFD)**, to quantify and analyze the generalization behavior of diffusion models, revealing key phenomena such as memorization-to-generalization transition and bias-variance tradeoffs.

Publications

C=Conference, J=Journal, S=In Submission, T=Thesis

[J.1] **Bayesian Optimization for Pareto Set Learning with Mixed Input**.
Weishi Wang, **Zijian Huang**, Wei Li, Hantang Qin, Judy Jin, Raed Al Kontar.
Ongoing.

[J.2] **ParetoTransfer: Transfer Learning for Multi-Objective Bayesian Optimization**.
Zijian Huang, Tianhan Gao, David Fenning, Chinedum Okwudire, Neil Dasgupta, Wei Lu, and Raed Al Kontar.
Ongoing.

[C.1] **Understanding Generalization in Diffusion Models via Probability Flow Distance**.
Huijie Zhang, **Zijian Huang**, Siyi Chen, Jinfan Zhou, Zekai Zhang, Peng Wang, Qing Qu.
In submission to ICML 2026, a short version has been accepted as a poster at HiLD Workshop at ICML 2025 and DeepMath 2025.

Skills

- **Programming Languages:** Python, Matlab, C++, Verilog, Spice
- **Data Science & Machine Learning:** Pytorch, Sklearn
- **Mathematical & Statistical Tools:** Mathematica, Matlab, Excel
- **Languages:** Chinese (native), English (fluent), Japanese (basic), German (basic)

Honors and Awards

- **1st Place Prize, IOE Undergraduate Senior Design Expo** *May 2026*
  University of Michigan
- **Michigan Engineering Continuing Student Scholarship** *Jul. 2025*
  University of Michigan
- **Accenture Industrial and Operations Engineering Scholarship** *Mar. 2025*
  University of Michigan
- **Dean's List** *Feb. 2025*
  University of Michigan
- **Undergraduate Excellent Scholarship** *Nov. 2024*
  Shanghai Jiao Tong University
- **National Scholarship** *Nov. 2024*
  Ministry of Education of the People's Republic of China
- **Undergraduate Excellent Scholarship** *Nov. 2023*
  Shanghai Jiao Tong University
- **Yu liming Scholarship** *Oct. 2023*
  SJTU Joint Institute
- **Annual Excellent Volunteer** *Dec. 2022*
  SJTU Joint Institute

Talks

- **Bayesian Optimization for Pareto Set Learning with Mixed Input** *Oct 2025*
  INFORMS 2025
- **Sampling from a Distribution** *Aug 2025*
  UMich Math Directed Reading Program

Leadership Experience

- **Minister of Sport Department** *Sept. 2023 - Aug. 2024*
  Student Union of SJTU Joint Institute
  - Responsible for organizing sports related activities within the college
  - Organized dozens of events, many with over 100 participants

Teaching Experience

- **Teaching Assistant, MATH2860J (Ordinary Differential Equations)** *Aug. 2024 - Dec. 2024*
  SJTU Joint Institute
  - Organizing recitation classes and office hour for MATH2860J (Honors Mathematics IV Ordinary Differential Equations)
- **Teaching Assistant, MATH2850J (Linear Algebra and Functions of Multiple Variables)** *May. 2024 - Aug. 2024*
  SJTU Joint Institute
  - Organizing recitation classes and office hour, grading assignments and exams for MATH2850J (Honors Mathematics III Linear Algebra and Functions of Multiple Variables)